

*Attempt this Paper on Separate Answer Sheet provided.*

Attempt all questions.

- Q. No. 2. Write a short note on the following:
- International Standard Organization (ISO) 249 $\Rightarrow$ 2017
 - Attribute control charts $2017, 224$
 - Failure costs in TQM 2017
 - Principles of quality management.

(5 each)

- Q. No. 3. 2017 TQM State the various Obstacles for the implementation of TQM.

(07)

- Q. No. 4. Take a double sampling plan with $n_1 = 50, c_1 = 2, n_2 = 100, c_2 = 6$ where n_1 & n_2 are the sample sizes and c_1 and c_2 are the acceptance numbers. If the incoming lots have fraction nonconforming $p' = 0.05$ then what is the probability of acceptance on the first sample? What is the probability of final acceptance?

(10)

199

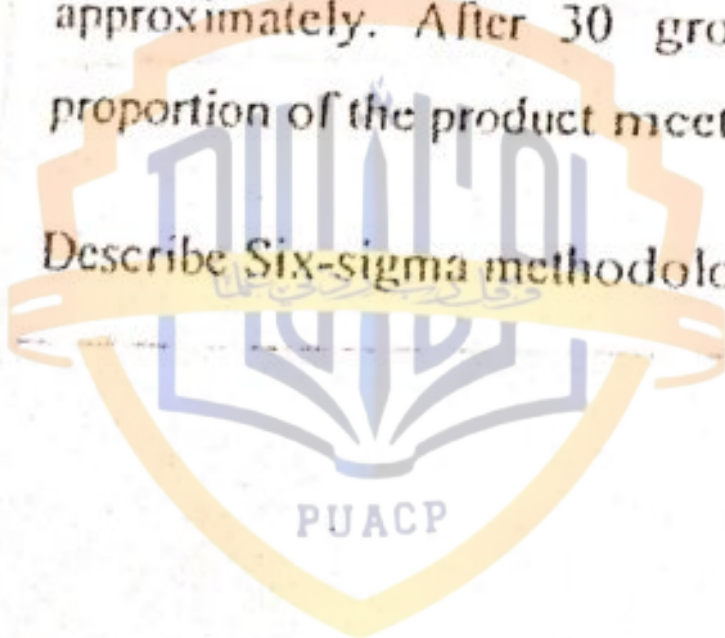
- Q. No. 5. Assuming that the distribution generated by the process described on the resistance of certain electrical parts with the subgroup size 25 is normal approximately. After 30 groups $\sum \bar{X} = 2419.2$ and $\sum \sigma = 107.6$, what proportion of the product meets the specifications 85 ± 10 ? ✓

(06)

p 21

- Q. No. 6. Describe Six-sigma methodology? Also describe the history of six-sigma.

(07)



$P = \binom{n}{d_1} p^{d_1} q^{n-d_1}$
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UNIVERSITY OF THE PUNJAB

Eighth Semester - 2017

Examination: B.S. 4 Years Programme

Roll No.

PAPER: Total Quality Management
Course Code: STAT-415

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

Attempt all questions.

- Q. No. 2. Write a short note on the following: (5 each)
- Total Quality Management
 - Competitive benchmarking
 - International Standard Organization (ISO)
 - Failure costs in TQM
- Q. No. 3. Describe Deming's Fourteen points for quality improvement in detail. (07)
- Q. No. 4. Take a sampling plan with $n_1 = 50$, $c_1 = 1$, $n_1 + n_2 = 150$, $c_2 = 4$ (08)
If the incoming lots have fraction nonconforming $p = 0.05$ then what is the probability of final acceptance? Calculate the probability of rejection on the first sample?
Where n_1 & n_2 are the sample sizes and c_1 and c_2 are the acceptance numbers.
- Q. No. 5. An item is made in lots of 200 each. The lots are given 100% inspection. The record sheet for the first 25 lots inspected showed that a total of 75 items were defective. (07)
- Determine the trial control limits for an np-chart showing number of defectives in each lot.
 - Assume that all points fall within the control limits. What is your estimate of the process average fraction defective p' ?
- If this p' remains unchanged. What is the probability that (i) the 26th lot will contain exactly 7 defectives? (ii) it will contain 7 or more defectives?
- Q. No. 6. What is benchmarking? Also describe the advantages and disadvantages of benchmarking? (08)





UNIVERSITY OF THE PUNJAB

Eighth Semester – 2018
Examination: B.S. 4 Years Program

Roll No. 010606

PAPER: Total Quality Management
Course Code: STAT-415 Part – II

MAX. TIME: 2 Hrs. 45 Min.
MAX. MARKS: 50

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q. No. 2. Write a short note on the following:

(5 each)

- Dimensions of Quality
- Eras of quality management
- Objectives of ISO
- External Failure costs in TQM

Q. No. 3. Describe Crosby's Fourteen points for quality improvement in detail.

(07)

Q. No. 4. A double sampling plan is as follows:

(08)

Select a sample of 2 articles from a lot of 20 articles. If both articles inspected are good, accept the lot. If both are defective, reject the lot. If one is good and one is defective, take a second sample of one article. If the article in the second sample is good, accept the lot. If it is defective, reject the lot.

If a lot having 25% defectives is submitted, what is the probability of acceptance?

Q. No. 5. Samples each of size $n=6$ items are taken from a manufacturing process at regular intervals. A normally distributed quality characteristic is measured and $\bar{x}$ and S values are calculated for each sample. After analyzing 50 samples, we have

(10)

$$\bar{\bar{x}} = 1000$$

$$\sum_{i=1}^{50} \bar{x}_i = 10000, \sum_{i=1}^{50} S_i = 75$$

- Compute the control limits for the $\bar{x}$ and S control charts.
- Assume that all points on both charts plot within the control limits. What are the natural tolerance limits of the process?
- If the specification limits are 19 ± 4.0 , what are your conclusions regarding the ability of the process to produce items conforming to specifications?
- Assuming that if an item exceeds the upper specification limit it can be reworked, and if it is below the lower specification limit it must be scrapped, what percent scrap and rework is the process now producing?
- Make suggestions as to how the process performance could be improved.

Q. No. 6. What is acceptance sampling? Discuss major types of acceptance sampling plans.

(05)



ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q. No. 2. Write a short note on the following:

(5 each)

- i. Six Sigma Methodology
- ii. Philosophy of TQM
- iii. Acceptance Sampling Plans → 140.
- iv. Internal Failure costs in TQM

Q. No. 3. Elaborate Deming's Fourteen Points to implement TQM.

(08)

Q. No. 4. What are the obstacles to implementing TQM? Also differentiate TQM with traditional approach.

(08)

Q. No. 5. The following table refers to the production and number of defectives of 12 lots with a constant lot size 200.

(06)

Lot Number	Number of Defectives	Lot Number	Number of Defectives
1	23	7	25
2	15	8	31
3	17	9	29
4	15	10	0
5	41	11	8
6	0	12	16

- a. Compute 3-sigma control limits for the fraction defectives.
- b. Plot the data on a control chart and make a decision about rejected lots.
Make revised control limits if necessary.

Q. No. 6. Why an organization should implement ISO? Also describe the internal and external objectives of this organization. → 253
→ 260 → 261

(08)



ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2. Answer the following short questions.

(4x5=20)

- Process Capability
- Eras of quality management
- Rectifying inspection - 174
- External Failure costs in TQM

Answer the following questions.

Q. No. 3. Elaborate Crosby's Fourteen Points to implement TQM.

(07)

Q. No. 4. Samples of $n=6$ items are taken from a manufacturing process at regular intervals. A normally distributed quality characteristic is measured and $\bar{x}$ and S values are calculated for each sample. After analyzing 50 groups, we have

(10)

$$\sum_{i=1}^{50} \bar{x}_i = 1000, \sum_{i=1}^{50} S_i = 75 \rightarrow P-51$$

1. Compute the control limits for the $\bar{x}$ and S control charts.
2. Assume that all points on both charts plot within the control limits. What are the natural tolerance limits of the process?
3. If the specification limits are 19 ± 4.0 , what are your conclusions regarding the ability of the process to produce items conforming to specifications?
4. Assuming that if an item exceeds the upper specification limit it can be reworked, and if it is below the lower specification limit it must be scrapped, what percent scrap and rework is the process now producing?
5. Make suggestions as to how the process performance could be improved.

Q. No. 5. Take a sampling plan with $n_1 = 50, c_1 = 1, n_2 = 100, c_2 = 4$

(08)

If the incoming lots have fraction nonconforming $p = 0.05$ then what is the probability of acceptance on the first sample? What is the probability of final acceptance? Calculate the probability of rejection on the first sample?

Q. No. 6. What is acceptance sampling? Discuss major advantages of acceptance sampling plans.

(05)

P-140

PUACP



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Eighth Semester – Spring 2022

Paper: Total Quality Management

Course Code: STAT-415

Roll No. 26428...

Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Write a short note on the following.

(6x5=30)

- Statistical quality control
- Waring limits
- Natural tolerance limits
- Average Outgoing Quality
- Process Capability
- Producer's and Consumer's risk

Write detailed answers to the following questions.

(3x10=30)

Q. No. 2. Why an organization should implement ISO? Also describe the internal and external objective of this organization.

Q. No. 3. Samples of $n = 7$ items are taken from a process at regular intervals. A normally distributed quality characteristic is measured and $\bar{x}$ and R values are calculated for each sample. After 35 subgroups have been analyzed, we have

$$\sum_{i=1}^{35} \bar{x}_i = 7805 \text{ and } \sum_{i=1}^{35} R_i = 1200$$

- Compute the control limit for the $\bar{x}$ and R control charts.
- Assume that all points on both charts plot within the control limits. What are the natural tolerance limits of the process?
- If the specification limits are 220 ± 35 , what are your conclusions regarding the ability of the process to produce items conforming to specifications?
- Assuming that if an item exceeds the upper specification limit it can be reworked, and if it is below the lower specification limit it must be scrapped, what percent scrap and rework is the process now producing?
- If the process were centered at $\mu = 220$, what would be the effect on percent scrap and rework?

Q. No. 4. Suppose that a supplier ships components in lots of size 5000. A single-sampling plan with $n = 50$ and $c = 2$ is being used for receiving inspection. Rejected lots are screened, and all defective items are reworked and returned to the lot. Draw the OC curve for this plan.



THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions.

(6x5=30)

- Total Quality Management
- Specification limits
- Natural tolerance limits
- Advantages of Double Sampling Plan
- Six-sigma methodology
- Rectifying Inspection

Answer the following questions.

(3x10=30)

Q. No. 2. What is benchmarking? Also describe the advantages and disadvantages of benchmarking.

Q. No. 3. Samples of $n = 4$ items are taken from a process at regular intervals. A normally distributed quality characteristic is measured and $\bar{x}$ and s values are calculated for each sample. After 50 subgroups have been analyzed, we have

$$\sum_{i=1}^{50} \bar{x}_i = 1000 \text{ and } \sum_{i=1}^{50} s_i = 72$$

- Compute the control limit for the $\bar{x}$ and s control charts
- Assume that all points on both charts plot within the control limits. What are the natural tolerance limits of the process?
- If the specification limits are 19 ± 4.0 , what are your conclusions regarding the ability of the process to produce items conforming to specifications?
- Assuming that if an item exceeds the upper specification limit it can be reworked, and if it is below the lower specification limit it must be scrapped, what percent scrap and rework is the process now producing?
- If the process were centered at $\mu = 19$, what would be the effect on percent scrap and rework?

Q. No. 4. The following table refers to the production and number of defectives of 12 lots with a constant lot size 200.

Lot Number	Number of Defectives	Lot Number	Number of Defectives
1	23	7	25
2	15	8	31
3	17	9	29
4	15	10	0
5	41	11	8
6	0	12	16

- Compute 3-sigma control limits for the fraction defectives.
- Plot the data on a control chart and make a decision about rejected lots.
- Make revised control limits if necessary.